

Lemma. Let $X: \mathcal{U} \subset \mathbb{R}^2 \rightarrow S$ be a parametrization of a regular surface S .
Given $q \in \mathcal{U}$,

$$DX_q[\mathbb{R}^2] = \{ \alpha'(0) \mid \alpha: (-\varepsilon, \varepsilon) \rightarrow S \text{ is differentiable and } \alpha(0) = X(q) \}.$$

Proof. Let w be a tangent vector at $X(q)$. That is,
for some smooth curve $\alpha: (-\varepsilon, \varepsilon) \rightarrow X[\mathcal{U}] \subset S$ such that $\alpha(0) = X(q)$, $w = \alpha'(0)$.
Since $X^{-1}: X[\mathcal{U}] \rightarrow \mathbb{R}^2$ is differentiable, the curve $\beta = X^{-1} \circ \alpha: (-\varepsilon, \varepsilon) \rightarrow \mathcal{U}$ is also differentiable, so $DX_q \cdot \beta'(0) = DX_q \cdot DX_{X(q)}^{-1} \cdot \alpha'(0) = \alpha'(0)$.

Conversely, let $w = DX_q(v)$ for some $v \in \mathbb{R}^2$. Then, a curve

$$\gamma: (-\varepsilon, \varepsilon) \rightarrow \mathcal{U}: t \mapsto t v + q$$

satisfies that $\gamma'(0) = v$ and $\gamma(0) = q$. Now, a curve $\alpha = X \circ \gamma$ satisfies $\alpha(0) = X(q)$ and $\alpha'(0) = DX_q \cdot v = w$, so proved. $\square$

Def. Let $S \subset \mathbb{R}^3$ be a regular surface and $p \in S$ be a point.
Then, for a parametrization $X: \mathcal{U} \rightarrow S$ at $p \in S$, a unit vector

$$N(q) = \frac{X_u \times X_v}{|X_u \times X_v|}(q) \quad \text{where } q = X^{-1}(p)$$

is called an unit normal vector at p .

Def. Given a regular surface S and a point $p \in S$, by definition, there is a parametrization $X: \mathcal{U} \rightarrow S$ containing p , ($p \in X[\mathcal{U}]$). Then, for $q = X^{-1}(p) \in \mathcal{U}$, the Jacobian $dX_q: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is linear with rank 2. The plane $dX_q[\mathbb{R}^2]$ is called a tangent plane at p for S , denote $T_p S$.

The tangent plane $T_p S$ is independent of choice of the parametrization X . But, given a parametrization X , $T_p S$ is determined by a basis $\left\{ \frac{\partial X}{\partial u}(q), \frac{\partial X}{\partial v}(q) \right\}$, it is denoted simply $\{X_u, X_v\}$.

Given a parametrization $X: \mathcal{U} \rightarrow S$ at $p \in S$, determine a vector $w \in T_p S$.

Given $w \in T_p S$, there is a curve $\alpha: (-\epsilon, \epsilon) \rightarrow X[\mathcal{U}]$ satisfying $\alpha(0) = p, \alpha'(0) = w$. Let $\beta = X^{-1} \circ \alpha: (-\epsilon, \epsilon) \rightarrow \mathcal{U}$. It is also smooth, and $\alpha = X \circ \beta$. Now,

$$\alpha'(0) = \frac{d}{dt}(X \circ \beta)(0) = \frac{d}{dt}X(u(t), v(t))(0) = X_u(q) \cdot u'(0) + X_v(q) \cdot v'(0)$$

where $q = X^{-1}(p)$ and $\beta(t) = (u(t), v(t))$.

Let S be a regular surface and $V \subseteq \mathbb{R}^3$ be an open set such that $S \subseteq V$.
If $f: V \rightarrow \mathbb{R}$ is differentiable, then for any point $p \in S$ and
parametrization $X: U \rightarrow S$ at $p \in S$, a map $f \circ X: U \rightarrow \mathbb{R}$ is differentiable.

Height function.

Given a surface S and a fixed unit vector $V \in \mathbb{R}^3$, a function

$h: S \rightarrow \mathbb{R}; p \mapsto \langle p, V \rangle$ is called a height function with respect to $V \in \mathbb{R}^3$.

Unit Sphere.

Let $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ be given. Define a map

$$R_{z, \theta}: \mathbb{R}^3 \rightarrow \mathbb{R}^3; (x, y, z) \mapsto (\cos \theta \cdot x - \sin \theta \cdot y, \sin \theta \cdot x + \cos \theta \cdot y, z).$$

(That is, Z -rotation of degree θ). Since $\cos \theta, \sin \theta$ are just constant, $R_{z, \theta}$ is differentiable. So, the restriction $R_{z, \theta}|_{S^2}$ is also differentiable.

Calculate the tangent plane at a point $p \in S^2$. Simply, denote $R = R_{z, \theta}$. Since S^2 is regular surface, there is a smooth curve $\alpha: (-\epsilon, \epsilon) \rightarrow S^2$ such that $\alpha(0) = p$.

Problems, do Comma sec 2.4.

Explicit the tangent plane of $x^2 + y^2 - z^2 = 1$ at $(x_0, y_0, 0)$.

Suppose that $x_0, y_0 > 0$. Consider a patch

$$X: D \rightarrow S: (u, v) \mapsto (\sqrt{1-u^2+v^2}, u, v)$$

where D is unit disk.

